

$P_1 = 604,86^7 \text{ g}$ $P_{1\text{change}} = 614,70 \text{ g}$
 $P_2 = 605,50 \text{ g}$ $P_{2\text{change}} = 614,62 \text{ g}$

HighSTEPS Experiment

Exp. Name: h333
Operator: Arthur + Barnaby

Date: 23-6-22

Sample Holders Thickness w/o Gouge or Bare Surface:

☒ Simple Grooves (3.4 cm x 3.4 cm): 34 x 34 mm
☐ Small Bare Surface (3.4 cm x 2.0 cm): _____ mm
☐ Large Bare Surface (3.4 cm x 4.0 cm): _____ mm
☐ Vessel Bare Surface (3.4 cm x 2 cm): _____ mm
☐ Porous Frits: _____ mm

Spacers: 1 (thickness) mm

Total Thickness-Bench: _____ mm

Material (Quartz, Granite, ...?): altered basalt

☒ Gouge- Particle Size, Size Distribution: < 125 μm
☐ Bare Surface - Roughness, Grit: _____

RN 1713 D = 2806 mm Box 4

Vert. Load Initial Value (kN): 0 kN

Horz. Load Initial Value (kN): -0,17 kN

Horizontal Load PID Setting:

gain: 0.003

Ti: 0.02

Td: 0.001

Normal Stress(es): 11.560 (kN) 10 (MPa)

Confining Pressure: / (MPa)

Pore Pressure SX: / (MPa)

Pore Pressure DX: / (MPa)

Sliding Velocity: 10 $\mu\text{m s}^{-1}$

Acceleration: _____

Deceleration: _____

Starting Position: _____

Shear Displ. Sample: _____

Velocity Steps: 3 - 10 - 30 - 100 - 300

Holds: 1 - 3 - 10 - 30 - 100 - 300 - 1000 - 3000

Temperature ($^{\circ}\text{C}$): RT

Relative Humidity (%): sec

Comments:

thinness square

605.53g = P_I
604.96g = P_{II}

$P_{Ic} = 615.48g$
 $P_{IIc} = 614.68g$

HighSTEPS Experiment

Exp. Name: h347
Operator: Florian

Date: 29.7.22

Sample Holders Thickness w/o Gouge or Bare Surface:

☒ Simple Grooves (3.4 cm x 3.4 cm): 34 x 34 mm
☐ Small Bare Surface (3.4 cm x 2.0 cm): _____ mm
☐ Large Bare Surface (3.4 cm x 4.0 cm): _____ mm
☐ Vessel Bare Surface (3.4 cm x 2 cm): _____ mm
☐ Porous Frits: _____ mm

Spacers: thinnest _____ mm

Total Thickness-Bench: _____ mm

Material (Quartz, Granite, ...?): BN-17B D=2806m basalt altered

☒ Gouge - Particle Size, Size Distribution: < 125 μm
☐ Bare Surface - Roughness, Grit: _____

Vert. Load Initial Value (kN): -0.07
Horz. Load Initial Value (kN): -0.21

Horizontal Load PID Setting:

gain: 0.003
Ti: 0.02
Td: 0.001

Normal Stress(es): 11.56 (kN) 6 (MPa)

Confining Pressure: _____ (MPa)
Pore Pressure SX: _____ (MPa)
Pore Pressure DX: _____ (MPa)

Sliding Velocity: 10 $\mu m/s$

Acceleration: $10^6 (5 \cdot 10^6) \mu m/s^2$
Deceleration: $10^6 (5 \cdot 10^6) \mu m/s^2$

Starting Position: _____

Shear Displ. Sample: _____

Velocity Steps: 10 - 3 - 10 - 20 - 100 - 300

Holds: 1-3-10-30-100-300-1000-3000

Temperature (°C): RT

Relative Humidity (%): wet

Comments:

thinnest square for sample prep

wetting: 7h57 - 8h37

cooler broke down during last hold

605.48
604.85

615.27
614.76

HighSTEPS Experiment

Exp. Name: h356

Date: 25.8.22

Operator: Florian

Sample Holders Thickness w/o Gouge or Bare Surface:

☒ Simple Grooves (3.4 cm x 3.4 cm): 34x34 mm

☐ Small Bare Surface (3.4 cm x 2.0 cm): _____ mm

☐ Large Bare Surface (3.4 cm x 4.0 cm): _____ mm

☐ Vessel Bare Surface (3.4 cm x 2 cm): _____ mm

☐ Porous Frits: _____ mm

Spacers: _____ mm

Total Thickness-Bench: _____ mm

Material (Quartz, Granite, ...?): R2806 basalt RN-1713

☒ Gouge - Particle Size, Size Distribution: <125 μ m

☐ Bare Surface - Roughness, Grit: _____

Vert. Load Initial Value (kN): 0.09

Horz. Load Initial Value (kN): -0.22

Horizontal Load PID Setting:

gain: 0.008

Ti: 0.02

Td: 0.001

Normal Stress(es): 34.68 (kN) 70 (MPa)

Confining Pressure: _____ (MPa)

Pore Pressure SX: _____ (MPa)

Pore Pressure DX: _____ (MPa)

Sliding Velocity: 10 μ m/s

Acceleration: 6' (5.10°) P75%

Deceleration: "

Starting Position: _____

Shear Displ. Sample: _____

Velocity Steps: 10 - 3 - 1.5 - 70 - 100 - 700

Holds: 1 - 3 - 10 - 70 - 100 - 700 - 1000 - 3000

Temperature (°C): RT

Relative Humidity (%): RH

Comments:

thinnest square for sample preparation

I 604,87 614,54
II 605,52 615,57

HighSTEPS Experiment

Exp. Name: h351
Operator: Floian - Arthur

Date: 12.8. 12.8.22

Sample Holders Thickness w/o Gouge or Bare Surface:

___ Simple Grooves (3.4 cm x 3.4 cm): 34 x 34 mm
___ Small Bare Surface (3.4 cm x 2.0 cm): _____ mm
___ Large Bare Surface (3.4 cm x 4.0 cm): _____ mm
___ Vessel Bare Surface (3.4 cm x 2 cm): _____ mm
___ Porous Frits: _____ mm

Spacers: thinnest mm

Total Thickness-Bench: _____ mm

Material (Quartz, Granite, ...?): RN-17B, D=2806 u Box 4 altered basalt

☒ Gouge - Particle Size, Size Distribution: < 125 μ m

___ Bare Surface - Roughness, Grit: _____

Vert. Load Initial Value (kN): 0

Horz. Load Initial Value (kN): -0,31

Horizontal Load PID Setting:

gain: _____

Ti: _____

Td: _____

Normal Stress(es): 34,68 (kN) 30 (MPa)

Confining Pressure: _____ (MPa)

Pore Pressure SX: _____ (MPa)

Pore Pressure DX: _____ (MPa)

Sliding Velocity: 10 μ m.s⁻¹

Acceleration: _____

Deceleration: _____

Starting Position: _____

Shear Displ. Sample: _____

Velocity Steps: 10-3-10-30-100-300

Holds: 1-3-10-30-100-300-1000-3000

Temperature (°C): RT

Relative Humidity (%): Wet

Comments:

~~11h27~~ - 11h27 - 12h10 (saturation)

- thinnest square

606.14g 615.41g
606.58g 615.97g

HighSTEPS Experiment

Exp. Name: h429
Operator: Burkley

Date: 1-11-22

Sample Holders Thickness w/o Gouge or Bare Surface:

☒ Simple Grooves (3.4 cm x 3.4 cm): 34x34 mm
☐ Small Bare Surface (3.4 cm x 2.0 cm): _____ mm
☐ Large Bare Surface (3.4 cm x 4.0 cm): _____ mm
☐ Vessel Bare Surface (3.4 cm x 2 cm): _____ mm
☐ Porous Frits: _____ mm

Spacers: thinnest mm

Total Thickness-Bench: _____ mm

Material (Quartz, Granite, ...?): R2806

☒ Gouge - Particle Size, Size Distribution: 425 μ m
☐ Bare Surface - Roughness, Grit: _____

Vert. Load Initial Value (kN): -0.2
Horz. Load Initial Value (kN): -0.62

Horizontal Load PID Setting:

gain: 0.003
Ti: 0.02
Td: 0.001

Normal Stress(es): 57.8 (kN) 50 (MPa)

Confining Pressure: _____ (MPa)
Pore Pressure SX: _____ (MPa)
Pore Pressure DX: _____ (MPa)

Sliding Velocity: 10 μ m/s for 11mm

Acceleration: 506 μ m/s²
Deceleration: _____

Starting Position: _____

Shear Displ. Sample: _____

Velocity Steps: 10-3-10-30-100-300
Holds: 1-3-10-30-100-300-1000-3000

Temperature (°C): RT
Relative Humidity (%): dry

Comments:

thinnest spacer used for setup

605.50g 614.58g
604.84g 613.97g

HighSTEPS Experiment

Exp. Name: h430
Operator: Barnaby

Date: 2-11-22

Sample Holders Thickness w/o Gouge or Bare Surface:

☒ Simple Grooves (3.4 cm x 3.4 cm): 34 x 34 mm
☐ Small Bare Surface (3.4 cm x 2.0 cm): _____ mm
☐ Large Bare Surface (3.4 cm x 4.0 cm): _____ mm
☐ Vessel Bare Surface (3.4 cm x 2 cm): _____ mm
☐ Porous Frits: _____ mm

Spacers: thinnest mm

Total Thickness-Bench: _____ mm

Material (Quartz, Granite, ...?): 22806

☒ Gouge - Particle Size, Size Distribution: 425 μ m
☐ Bare Surface - Roughness, Grit: _____

Vert. Load Initial Value (kN): -0.1
Horz. Load Initial Value (kN): -0.47

Horizontal Load PID Setting:

gain: 0.03
Ti: 0.02
Td: 0.001

Normal Stress(es): 57.8 (kN) 50 (MPa)

Confining Pressure: _____ (MPa)

Pore Pressure SX: _____ (MPa)

Pore Pressure DX: _____ (MPa)

Sliding Velocity: 10 μ m/s for 11mm

Acceleration: 5e6 μ m/s²
Deceleration: _____

Starting Position: _____

Shear Displ. Sample: _____

Velocity Steps: 10-3-10-30-100-300

Holds: 1-3-10-30-100-300-1000-3000

Temperature (°C): RT

Relative Humidity (%): Wet

Comments:

thinnest square used for setup
saturated @ 1 MPa 8:26 - 9:06

sample holders 604,88 615,18
605,48 615,99

HighSTEPS Experiment

Exp. Name: h329
Operator: Florian

Date: 14.6.22

Sample Holders Thickness w/o Gouge or Bare Surface:

☒ Simple Grooves (3.4 cm x 3.4 cm): 74x37 mm
☐ Small Bare Surface (3.4 cm x 2.0 cm): _____ mm
☐ Large Bare Surface (3.4 cm x 4.0 cm): _____ mm
☐ Vessel Bare Surface (3.4 cm x 2 cm): _____ mm
☐ Porous Frits: _____ mm

Spacers: thinest mm

Total Thickness-Bench: _____ mm

Material (Quartz, Granite, ...?): R4641, IDPP-2 core 11 1302

☒ Gouge - Particle Size, Size Distribution: < 125 μ m
☐ Bare Surface - Roughness, Grit: _____

Vert. Load Initial Value (kN): -0.2

Horz. Load Initial Value (kN): -0.4

Horizontal Load PID Setting:

gain: 0.003

Ti: 0.02

Td: 0.001

Normal Stress(es): 11.56 (kN) 10 (MPa)

Confining Pressure: / (MPa)

Pore Pressure SX: / (MPa)

Pore Pressure DX: / (MPa)

Sliding Velocity: 10 μ m/s

Acceleration: _____

Deceleration: _____

Starting Position: _____

Shear Displ. Sample: _____

Velocity Steps: 10 - 3 - 10 - 30 - 100 - 300

Holds: 1 - 3 - 10 - 30 - 100 - 300 - 1000 - 3000

Temperature (°C): RT

Relative Humidity (%): RH

Comments:

thinest square for sample prep

sample holders 604,88 614,89
605,50 615,72

HighSTEPS Experiment

Exp. Name: h 332

Date: 15.6.22

Operator: Florian

Sample Holders Thickness w/o Gouge or Bare Surface:

☒ Simple Grooves (3.4 cm x 3.4 cm): 34 x 34 mm

☐ Small Bare Surface (3.4 cm x 2.0 cm): mm

☐ Large Bare Surface (3.4 cm x 4.0 cm): mm

☐ Vessel Bare Surface (3.4 cm x 2 cm): mm

☐ Porous Frits: mm

Spacers: thinnest mm

Total Thickness-Bench: mm

Material (Quartz, Granite, ...?): R4641, 1DDP-2, Core 1, Box 5 D=4641 μ

☒ Gouge - Particle Size, Size Distribution: < 125 μ

☐ Bare Surface - Roughness, Grit:

Vert. Load Initial Value (kN): -0.2

Horz. Load Initial Value (kN): -0.4

Horizontal Load PID Setting:

gain: 0.003

Ti: 0.02

Td: 0.001

Normal Stress(es): 11.56 (kN) 10 (MPa)

Confining Pressure: / (MPa)

Pore Pressure SX: / (MPa)

Pore Pressure DX: / (MPa)

Sliding Velocity: 10 μ/s

Acceleration:

Deceleration:

Starting Position:

Shear Displ. Sample:

Velocity Steps: 10 - 3 - 10 - 30 - 100 - 300

Holds: 1 - 3 - 10 - 30 - 100 - 300 - 1000 - 3000

Temperature (°C): RT

Relative Humidity (%): wet

Comments:

thinnest square for sample prep
walking 11:01 - 11:41

I = 605,56 g $T_c = 615,14$ g
II = 604,57 g $T_c = 614,68$ g

HighSTEPS Experiment

Exp. Name: h342
Operator: Brian Arthur

Date: 28.7.22

Sample Holders Thickness w/o Gouge or Bare Surface:

☒ Simple Grooves (3.4 cm x 3.4 cm): 34x34 mm
☐ Small Bare Surface (3.4 cm x 2.0 cm): _____ mm
☐ Large Bare Surface (3.4 cm x 4.0 cm): _____ mm
☐ Vessel Bare Surface (3.4 cm x 2 cm): _____ mm
☐ Porous Frits: _____ mm

Spacers: thinnest mm

Total Thickness-Bench: _____ mm

Material (Quartz, Granite, ...?): IDDP-2 D=4641 Box 5 Core 11

☐ Gouge - Particle Size, Size Distribution: <125 μ m
☐ Bare Surface - Roughness, Grit: _____

Vert. Load Initial Value (kN): -0,14

Horz. Load Initial Value (kN): -0,38

Horizontal Load PID Setting:

gain: 0,003

Ti: 0,02

Td: 0,002

Normal Stress(es): 34,68 (kN) 30 (MPa)

Confining Pressure: _____ (MPa)

Pore Pressure SX: _____ (MPa)

Pore Pressure DX: _____ (MPa)

Sliding Velocity: 10 μ m/s

Acceleration: _____

Deceleration: _____

Starting Position: _____

Shear Displ. Sample: _____

Velocity Steps: 10 - 3 - 10 - 30 - 100 - 300

Holds: 1 - 3 - 10 - 30 - 100 - 300 - 1000 - 3000

Temperature ($^{\circ}$ C): RT

Relative Humidity (%): dry - RH

Comments: thinnest square for sample prep

$P_I = 605,57 \text{ g}$ $P_{Ic} = 615,05 \text{ g}$
 $P_{II} = 604,91 \text{ g}$ $P_{IIc} = 614,93 \text{ g}$

HighSTEPS Experiment

Exp. Name: H341
Operator: Floian-Arthur

Date: 28/07/2022

Sample Holders Thickness w/o Gouge or Bare Surface:

☒ Simple Grooves (3.4 cm x 3.4 cm): 34x34 mm
☐ Small Bare Surface (3.4 cm x 2.0 cm): _____ mm
☐ Large Bare Surface (3.4 cm x 4.0 cm): _____ mm
☐ Vessel Bare Surface (3.4 cm x 2 cm): _____ mm
☐ Porous Frits: _____ mm

Spacers: thinnest mm

Total Thickness-Bench: _____ mm

Material (Quartz, Granite, ...?): IDPP-2 D=4641 m D₀ x 5 Core II

☒ Gouge - Particle Size, Size Distribution: < 125 μm
☐ Bare Surface - Roughness, Grit: _____

Vert. Load Initial Value (kN): -0.05

Horz. Load Initial Value (kN): -0.22

Horizontal Load PID Setting:

gain: 0.003

Ti: 0.02

Td: 0.001

Normal Stress(es): 34.68 (kN) 70 (MPa)

Confining Pressure: _____ (MPa)

Pore Pressure SX: _____ (MPa)

Pore Pressure DX: _____ (MPa)

Sliding Velocity: 10 $\mu\text{m/s}$

Acceleration: $10^{16} / 5 \cdot 10^{16} \mu\text{m/s}^2$

Deceleration: $10^{16} / 5 \cdot 10^{16} \mu\text{m/s}^2$

Starting Position: _____

Shear Displ. Sample: _____

Velocity Steps: 10 - 30 - 60 - 30 - 100 - 700

Holds: 1 - 3 - 10 - 70 - 100 - 300 - 1000 - 3000

Temperature (°C): RT

Relative Humidity (%): wet

Comments:

thinnest square for sample prep
wetting: 846 - 8451

606.15g 615.94g
606.56g 616.35g

HighSTEPS Experiment

Exp. Name: h413
Operator: Barnaby

Date: 20-10-22

Sample Holders Thickness w/o Gouge or Bare Surface:

☒ Simple Grooves (3.4 cm x 3.4 cm): 34 x 34 mm
☐ Small Bare Surface (3.4 cm x 2.0 cm): _____ mm
☐ Large Bare Surface (3.4 cm x 4.0 cm): _____ mm
☐ Vessel Bare Surface (3.4 cm x 2 cm): _____ mm
☐ Porous Frits: _____ mm

Spacers: thinner mm

Total Thickness-Bench: _____ mm

Material (Quartz, Granite, ...?): R4641

☒ Gouge - Particle Size, Size Distribution: <125 μ m
☐ Bare Surface - Roughness, Grit: _____

Vert. Load Initial Value (kN): -0.24
Horz. Load Initial Value (kN): -0.5

Horizontal Load PID Setting:

gain: 0.03
Ti: 0.02
Td: 0.001

Normal Stress(es): 57.8 (kN) 50 (MPa)

Confining Pressure: _____ (MPa)
Pore Pressure SX: _____ (MPa)
Pore Pressure DX: _____ (MPa)

Sliding Velocity: 10 μ m/s for 11mm

Acceleration: 56 μ m/s²
Deceleration: _____

Starting Position: _____

Shear Displ. Sample: _____

Velocity Steps: 10 - 7 - 6 - 30 - 100 - 300
Holds: 1 - 3 - 10 - 30 - 100 - 300 - 1000 - 3000

Temperature (°C): RT
Relative Humidity (%): dry

Comments:

thinner square used for setup

605.52g 615.57g
604.88g 614.88g

HighSTEPS Experiment

Exp. Name: h4/4
Operator: Barnaby

Date: 21-10-22

Sample Holders Thickness w/o Gouge or Bare Surface:

☒ Simple Grooves (3.4 cm x 3.4 cm): 34x34 mm
☐ Small Bare Surface (3.4 cm x 2.0 cm): _____ mm
☐ Large Bare Surface (3.4 cm x 4.0 cm): _____ mm
☐ Vessel Bare Surface (3.4 cm x 2 cm): _____ mm
☐ Porous Frits: _____ mm

Spacers: thinnest mm

Total Thickness-Bench: _____ mm

Material (Quartz, Granite, ...?): R4641

☒ Gouge - Particle Size, Size Distribution: <125 μ m
☐ Bare Surface - Roughness, Grit: _____

Vert. Load Initial Value (kN): -0.05
Horz. Load Initial Value (kN): -0.3

Horizontal Load PID Setting:

gain: 0.003
Ti: 0.02
Td: 0.001

Normal Stress(es): 101.65 57.8 (kN) 50 (MPa)

Confining Pressure: _____ (MPa)
Pore Pressure SX: _____ (MPa)
Pore Pressure DX: _____ (MPa)

Sliding Velocity: 10 μ m/s for 11 mm

Acceleration: 566 μ m/s²
Deceleration: _____

Starting Position: _____

Shear Displ. Sample: _____

Velocity Steps: 10-3-10-30-100-200
Holds: 1-3-10-30-100-300-1000-3000

Temperature (°C): RT
Relative Humidity (%): wet

Comments:

thinnest spacer used for setup
Setbacked @ 1MPa 9:32 - 10:12

sample holder 605, 56 616, 12
604, 98 615, 63

HighSTEPS Experiment

Exp. Name: h 316
Operator: Florida

Date: 4.5.5.22

Sample Holders Thickness w/o Gouge or Bare Surface:

☒ Simple Grooves (3.4 cm x 3.4 cm): 74x74 mm
☐ Small Bare Surface (3.4 cm x 2.0 cm): _____ mm
☐ Large Bare Surface (3.4 cm x 4.0 cm): _____ mm
☐ Vessel Bare Surface (3.4 cm x 2 cm): _____ mm
☐ Porous Frits: _____ mm

Spacers: / _____ mm

Total Thickness-Bench: _____ mm

Material (Quartz, Granite, ...?): R4648 (IDPP-2 Cor 12 Box 7 D=4648mm)

☒ Gouge - Particle Size, Size Distribution: <125 μ m
☐ Bare Surface - Roughness, Grit: _____

Vert. Load Initial Value (kN): -0.25

Horz. Load Initial Value (kN): -0.58

Horizontal Load PID Setting:

gain: 0.003

Ti: 0.02

Td: 0.001

Normal Stress(es): 11.56 (kN) 10 (MPa)

Confining Pressure: _____ (MPa)

Pore Pressure SX: _____ (MPa)

Pore Pressure DX: _____ (MPa)

Sliding Velocity: 10 μ m/s

Acceleration: _____

Deceleration: _____

Starting Position: _____

Shear Displ. Sample: _____

Velocity Steps: (10) - 7 - 10 - 70 - 100 - 300

Holds: 1 - 7 - 10 - 70 - 100 - 300 - 1000 - 3000

Temperature (°C): RT

Relative Humidity (%): RH

Comments:

touchpoint 8,9 mm
1156 kN @ 9,4 mm
1156 kN @ 10,3 mm
14,4 21,7

sample thickness 604.86 615.79
605.50 616.11

HighSTEPS Experiment

Exp. Name: h325
Operator: Barnaby Florian

Date: 13.5.22

Sample Holders Thickness w/o Gouge or Bare Surface:

☒ Simple Grooves (3.4 cm x 3.4 cm): 34 x 34 mm
Small Bare Surface (3.4 cm x 2.0 cm): _____ mm
Large Bare Surface (3.4 cm x 4.0 cm): _____ mm
Vessel Bare Surface (3.4 cm x 2 cm): _____ mm
Porous Frits: _____ mm

Spacers: thinnet mm

Total Thickness-Bench: _____ mm

Material (Quartz, Granite, ...?): R4648, 100P-2 G12 Box 4 D=4648

☒ Gouge - Particle Size, Size Distribution: < 125 μ m
Bare Surface - Roughness, Grit: _____

Vert. Load Initial Value (kN): -0.15

Horz. Load Initial Value (kN): -0.43

Horizontal Load PID Setting:

gain: 0.003

Ti: 0.02

Td: 0.001

Normal Stress(es): 11.56 (kN) 10 (MPa)

Confining Pressure: _____ (MPa)

Pore Pressure SX: _____ (MPa)

Pore Pressure DX: _____ (MPa)

Sliding Velocity: 10 μ m/s

Acceleration: 10⁶ (5 $\cdot$ 10⁶) μ m/s²

Deceleration: "

Starting Position: _____

Shear Displ. Sample: _____

Velocity Steps: 10 - 3 - 10 - 30 - 100 - 300

Holds: 1 - 3 - 10 - 30 - 100 - 300 - 1000 - 3000

Temperature (°C): RT

Relative Humidity (%): wet

Comments:

thinnet square for sample prep

shear pore noise at ~63265

wetting 11h24 - 12h04

HighSTEPS Experiment

Exp. Name: h 344
Operator: Florian

Date: 3.8.22

Sample Holders Thickness w/o Gouge or Bare Surface:

☒ Simple Grooves (3.4 cm x 3.4 cm): 34 x 34 mm
☐ Small Bare Surface (3.4 cm x 2.0 cm): _____ mm
☐ Large Bare Surface (3.4 cm x 4.0 cm): _____ mm
☐ Vessel Bare Surface (3.4 cm x 2 cm): _____ mm
☐ Porous Frits: _____ mm

Spacers: Minnest _____ mm

Total Thickness-Bench: _____ mm

Material (Quartz, Granite, ...?): 100P-2 Jotexh Core 12 13074 D = 4648m

☒ Gouge - Particle Size, Size Distribution: < 125 μ m
☐ Bare Surface - Roughness, Grit: _____

Vert. Load Initial Value (kN): -0.01

Horz. Load Initial Value (kN): -0.17

Horizontal Load PID Setting:

gain: 0.003

Ti: 0.02

Td: 0.001

Normal Stress(es): 34.68 (kN) 30 (MPa)

Confining Pressure: _____ (MPa)

Pore Pressure SX: _____ (MPa)

Pore Pressure DX: _____ (MPa)

Sliding Velocity: 10 μ m/s

Acceleration: 10⁶ (5.10⁶) μ m/s²

Deceleration: 10⁶ (5.10⁶) μ m/s²

Starting Position: _____

Shear Displ. Sample: _____

Velocity Steps: 10 - 2 - 10 - 30 - 100 - 300

Holds: 1 - 3 - 10 - 30 - 100 - 300 - 1000 - 3000

Temperature (°C): 127

Relative Humidity (%): RH

Comments:

Minnest square for sample prep. Arthur prepared
oscillations in σ_n at 9h30 for several minutes

605,48 615,78
604,86 615,15

HighSTEPS Experiment

Exp. Name: 6345
Operator: Florian

Date: 3.8.22

Sample Holders Thickness w/o Gouge or Bare Surface:

☒ Simple Grooves (3.4 cm x 3.4 cm): 74 x 74 mm
☐ Small Bare Surface (3.4 cm x 2.0 cm): _____ mm
☐ Large Bare Surface (3.4 cm x 4.0 cm): _____ mm
☐ Vessel Bare Surface (3.4 cm x 2 cm): _____ mm
☐ Porous Frits: _____ mm

Spacers: thinnest mm

Total Thickness-Bench: _____ mm

Material (Quartz, Granite, ...?): 100 P. 2 ddenit D = 4648 μ

☒ Gouge - Particle Size, Size Distribution: < 125 μm
☐ Bare Surface - Roughness, Grit: _____

Vert. Load Initial Value (kN): -0.11
Horz. Load Initial Value (kN): ~~0.44~~ -0.44

Horizontal Load PID Setting:
gain: 0.003
Ti: 0.02
Td: 0.001

Normal Stress(es): 34.68 (kN) 70 (MPa)

Confining Pressure: / (MPa)
Pore Pressure SX: / (MPa)
Pore Pressure DX: / (MPa)

Sliding Velocity: 10 P/s

Acceleration: 10⁶ (5-10⁶) P/s²
Deceleration: 10⁶ (5-10⁶) P/s²

Starting Position: /
Shear Displ. Sample: /

Velocity Steps: 10 - 3 - 10 - 30 - 100 - 700
Holds: 1 - 2 - 10 - 30 - 700 - 300 - 1000 - 7000

Temperature (°C): RT
Relative Humidity (%): wet

Comments:

thinnest square for sample prep
cetting: 14h⁰⁰ - 14h⁴⁰

606.12g 616.13g
606.52g 616.75g

HighSTEPS Experiment

Exp. Name: h415
Operator: Barinab

Date: 21-10-22

Sample Holders Thickness w/o Gouge or Bare Surface:

☒ Simple Grooves (3.4 cm x 3.4 cm): 34x34 mm
☐ Small Bare Surface (3.4 cm x 2.0 cm): _____ mm
☐ Large Bare Surface (3.4 cm x 4.0 cm): _____ mm
☐ Vessel Bare Surface (3.4 cm x 2 cm): _____ mm
☐ Porous Frits: _____ mm

Spacers: thinnest mm

Total Thickness-Bench: _____ mm

Material (Quartz, Granite, ...?): R4648

☒ Gouge - Particle Size, Size Distribution: <125 μ m
☐ Bare Surface - Roughness, Grit: _____

Vert. Load Initial Value (kN): -0.15
Horz. Load Initial Value (kN): -0.48

Horizontal Load PID Setting:

gain: 0.003
Ti: 0.2
Td: 0.001

Normal Stress(es): 57.8 (kN) 50 (MPa)

Confining Pressure: _____ (MPa)
Pore Pressure SX: _____ (MPa)
Pore Pressure DX: _____ (MPa)

Sliding Velocity: 10 μ m/s for 11mm

Acceleration: 5e6 μ m/s²
Deceleration: _____

Starting Position: _____

Shear Displ. Sample: _____

Velocity Steps: 10 - 3 - 10 - 30 - 100 - 300

Holds: 1 - 3 - 10 - 30 - 100 - 300 - 1000 - 3000

Temperature (°C): RT

Relative Humidity (%): dry

Comments:

thinnest square used for setup
magnetic?

605.51g 615.62g
604.86g 615.26g

HighSTEPS Experiment

Exp. Name: H416
Operator: Barnaby

Date: 24-10-22

Sample Holders Thickness w/o Gouge or Bare Surface:

☒ Simple Grooves (3.4 cm x 3.4 cm): 34 x 34 mm
☐ Small Bare Surface (3.4 cm x 2.0 cm): _____ mm
☐ Large Bare Surface (3.4 cm x 4.0 cm): _____ mm
☐ Vessel Bare Surface (3.4 cm x 2 cm): _____ mm
☐ Porous Frits: _____ mm

Spacers: thinnest mm

Total Thickness-Bench: _____ mm

Material (Quartz, Granite, ...?): R4648

☒ Gouge - Particle Size, Size Distribution: <125 μ m
☐ Bare Surface - Roughness, Grit: _____

Vert. Load Initial Value (kN): 0.05
Horz. Load Initial Value (kN): -0.16

Horizontal Load PID Setting:

gain: 0.003
Ti: 0.02
Td: 0.001

Normal Stress(es): 57.8 (kN) 50 (MPa)

Confining Pressure: _____ (MPa)
Pore Pressure SX: _____ (MPa)
Pore Pressure DX: _____ (MPa)

Sliding Velocity: 10 μ m/s Grillmm

Acceleration: 5e6 μ m/s²
Deceleration: _____

Starting Position: _____

Shear Displ. Sample: _____

Velocity Steps: 10-3-10-30-100-300
Holds: 1-3-10-30-100-300-1000-3000

Temperature (°C): RT
Relative Humidity (%): wet

Comments:

thinnest square used for setup
Saturated @ 1MPa 9:16-9:57

605,59 615,88
605,85 615,33

HighSTEPS Experiment

Exp. Name: h359
Operator: Fabian / Barnaby

Date: 26-8-22

Sample Holders Thickness w/o Gouge or Bare Surface:

☒ Simple Grooves (3.4 cm x 3.4 cm): 34 x 34 mm
☐ Small Bare Surface (3.4 cm x 2.0 cm): _____ mm
☐ Large Bare Surface (3.4 cm x 4.0 cm): _____ mm
☐ Vessel Bare Surface (3.4 cm x 2 cm): _____ mm
☐ Porous Frits: _____ mm

Spacers: Thinnest _____ mm

Total Thickness-Bench: _____ mm

Material (Quartz, Granite, ...?): R4654 Ddente IDDP-2 core13 D=4654m

☐ Gouge - Particle Size, Size Distribution: <125 μ m

☐ Bare Surface - Roughness, Grit: _____

Vert. Load Initial Value (kN): 0.025

Horz. Load Initial Value (kN): -0.35

Horizontal Load PID Setting:

gain: 0.003

Ti: 0.02

Td: 0.001

Normal Stress(es): 11.56 (kN) 10 (MPa)

Confining Pressure: _____ (MPa)

Pore Pressure SX: _____ (MPa)

Pore Pressure DX: _____ (MPa)

Sliding Velocity: 10 μ m/sec
11mm

Acceleration: 2.003

Deceleration: _____

Starting Position: _____

Shear Displ. Sample: _____

Velocity Steps: 3-10-30-100-300

Holds: 1-3-10-30-100-300-1000-3000

Temperature ($^{\circ}$ C): RT

Relative Humidity (%): Rel

Comments:

Thinnest square used for setup

read 1 MPa at 13kPa

605.46 615.73 g
604.85 615.21 g

HighSTEPS Experiment

Exp. Name: 4357

Date: 25-8-22

Operator: Bernady / Florian

Sample Holders Thickness w/o Gouge or Bare Surface:

☒ Simple Grooves (3.4 cm x 3.4 cm): 74x34 mm

☐ Small Bare Surface (3.4 cm x 2.0 cm): _____ mm

☐ Large Bare Surface (3.4 cm x 4.0 cm): _____ mm

☐ Vessel Bare Surface (3.4 cm x 2 cm): _____ mm

☐ Porous Frits: _____ mm

Spacers: thinner mm

Total Thickness-Bench: _____ mm

R4654

Material (Quartz, Granite, ...?): dolerite 100P-2 Core 13 D=4654u

☒ Gouge - Particle Size, Size Distribution: <125µm

☐ Bare Surface - Roughness, Grit: _____

Vert. Load Initial Value (kN): 0.02

Horz. Load Initial Value (kN): -0.42

Horizontal Load PID Setting:

gain: 0.005

Ti: 0.02

Td: 0.001

Normal Stress(es): 11.56 (kN) 10 (MPa)

Confining Pressure: _____ (MPa)

Pore Pressure SX: _____ (MPa)

Pore Pressure DX: _____ (MPa)

Sliding Velocity: 10 P/s

Acceleration: 6° (5.16°) P/s

Deceleration: "

Starting Position: _____

Shear Displ. Sample: _____

Velocity Steps: 10 - 3 - 10 - 30 - 100 - 300

Holds: 1-3-10-30-100-300-1000-3000

Temperature (°C): RT

Relative Humidity (%): Wet

Comments:

thinnest square for sample prep
saturation: 13h12 - 13h52
Held 10 MPa 14h09 - 1429

605.39 g → 615.42 g
604.81 g → 615.20 g

HighSTEPS Experiment

Exp. Name: h360
Operator: Barnaby / Florian

Date: 29.8.22

Sample Holders Thickness w/o Gouge or Bare Surface:

☒ Simple Grooves (3.4 cm x 3.4 cm): 74 x 74 mm
☐ Small Bare Surface (3.4 cm x 2.0 cm): _____ mm
☐ Large Bare Surface (3.4 cm x 4.0 cm): _____ mm
☐ Vessel Bare Surface (3.4 cm x 2 cm): _____ mm
☐ Porous Frits: _____ mm

Spacers: thinnest mm

Total Thickness-Bench: _____ mm

Material (Quartz, Granite, ...?): R4654

☒ Gouge - Particle Size, Size Distribution: <125 μ m
☐ Bare Surface - Roughness, Grit: _____

Vert. Load Initial Value (kN): 0.18

Horz. Load Initial Value (kN): -0.18

Horizontal Load PID Setting:

gain: 0.003

Ti: 0.02

Td: 0.001

Normal Stress(es): 34.68 (kN) 30 (MPa)

Confining Pressure: _____ (MPa)

Pore Pressure SX: _____ (MPa)

Pore Pressure DX: _____ (MPa)

Sliding Velocity: 10 P/s

Acceleration: 10⁶ (5-10⁶) P/s²

Deceleration: "

Starting Position: _____

Shear Displ. Sample: _____

Velocity Steps: 10 - 3 - 10 - 30 - 100 - 300

Holds: 1 - 3 - 10 - 30 - 100 - 300 - 1000 - 3000

Temperature (°C): RT

Relative Humidity (%): RH

Comments:

thinnest square for sample prep

1 MPa at 11:37

oscillations in normal force observed at 13414, 13450

I 605.53g 615.25g
II 604.94g 615.15g

HighSTEPS Experiment

Exp. Name: h358
Operator: laron/Bernady

Date: 26.8.22

Sample Holders Thickness w/o Gouge or Bare Surface:

☒ Simple Grooves (3.4 cm x 3.4 cm): 34x34 mm
☐ Small Bare Surface (3.4 cm x 2.0 cm): _____ mm
☐ Large Bare Surface (3.4 cm x 4.0 cm): _____ mm
☐ Vessel Bare Surface (3.4 cm x 2 cm): _____ mm
☐ Porous Frits: _____ mm

Spacers: (thin) mm

Total Thickness-Bench: _____ mm

Material (Quartz, Granite, ...?): R4654 Deterite IDDP-2 case B D=4654m

☒ Gouge - Particle Size, Size Distribution: <125 μ m
☐ Bare Surface - Roughness, Grit: _____

Vert. Load Initial Value (kN): 0.09

Horz. Load Initial Value (kN): -0.28

Horizontal Load PID Setting:

gain: 0.03

Ti: 0.02

Td: 0.001

Normal Stress(es): 34.68 (kN) 30 (MPa)

Confining Pressure: _____ (MPa)

Pore Pressure SX: _____ (MPa)

Pore Pressure DX: _____ (MPa)

Sliding Velocity: 10 μ m/sec

Acceleration: _____

Deceleration: _____

Starting Position: _____

Shear Displ. Sample: _____

Velocity Steps: 10-3-10-30-100-300

Holds: 1-3-10-30-100-300-1000-3000

Temperature (°C): RT

Relative Humidity (%): saturated

Comments:

Thinnest square for sample preparation
Melting: 8h10 - 8h50

run-in: 10h06

Leak on last SHS, refilled but temporarily exposed to air ~5min

606.15g 616.00g
606.57g 616.26

HighSTEPS Experiment

Exp. Name: h431
Operator: Burkeby

Date: 2-11-22

Sample Holders Thickness w/o Gouge or Bare Surface:

☒ Simple Grooves (3.4 cm x 3.4 cm): 34 x 34 mm
☐ Small Bare Surface (3.4 cm x 2.0 cm): _____ mm
☐ Large Bare Surface (3.4 cm x 4.0 cm): _____ mm
☐ Vessel Bare Surface (3.4 cm x 2 cm): _____ mm
☐ Porous Frits: _____ mm

Spacers: thinnest mm

Total Thickness-Bench: _____ mm

Material (Quartz, Granite, ...?): R4654

☒ Gouge - Particle Size, Size Distribution: 425 μ m

☐ Bare Surface - Roughness, Grit: _____

Vert. Load Initial Value (kN): -0.17

Horz. Load Initial Value (kN): -0.62

Horizontal Load PID Setting:

gain: 0.003

Ti: 0.02

Td: 0.001

Normal Stress(es): 57.8 (kN) 50 (MPa)

Confining Pressure: _____ (MPa)

Pore Pressure SX: _____ (MPa)

Pore Pressure DX: _____ (MPa)

Sliding Velocity: 0.1 mm/s for 11mm

Acceleration: 56 μ m/s²

Deceleration: _____

Starting Position: _____

Shear Displ. Sample: _____

Velocity Steps: 10-3-10-30-100-300

Holds: 1-3-10-30-100-300-1000-3000

Temperature (°C): RT

Relative Humidity (%): dry

Comments:

thinnest square used for setup

605.53g 615.44g
604.45g 614.87g

HighSTEPS Experiment

Exp. Name: h432
Operator: Barnaby

Date: 3-11-22

Sample Holders Thickness w/o Gouge or Bare Surface:

☒ Simple Grooves (3.4 cm x 3.4 cm): 34x34 mm
☐ Small Bare Surface (3.4 cm x 2.0 cm): _____ mm
☐ Large Bare Surface (3.4 cm x 4.0 cm): _____ mm
☐ Vessel Bare Surface (3.4 cm x 2 cm): _____ mm
☐ Porous Frits: _____ mm

Spacers: thinnest mm

Total Thickness-Bench: _____ mm

Material (Quartz, Granite, ...?): 24654
☒ Gouge - Particle Size, Size Distribution: < 125 μ m
☐ Bare Surface - Roughness, Grit: _____

Vert. Load Initial Value (kN): - 0.15
Horz. Load Initial Value (kN): - 0.45

Horizontal Load PID Setting:

gain: 0.003
Ti: 0.02
Td: 0.001

Normal Stress(es): 57.6 (kN) 50 (MPa)

Confining Pressure: _____ (MPa)
Pore Pressure SX: _____ (MPa)
Pore Pressure DX: _____ (MPa)

Sliding Velocity: 10 μ m/s for 11mm

Acceleration: 5e6 μ m/s²
Deceleration: _____

Starting Position: _____
Shear Displ. Sample: _____

Velocity Steps: 10-3-10-30-100-300
Holds: 1-3-10-30-100-300-1000-3000

Temperature (°C): RT
Relative Humidity (%): wet

Comments:

thinnest square used for step
magnetic?

Saturated @ 1MPa 8:32-9:13

sample holders 65,85 67,67
65,50 65,63

HighSTEPS Experiment

Exp. Name: h 326
Operator: Bernady Flusian

Date: 16.5.22

Sample Holders Thickness w/o Gouge or Bare Surface:

☒ Simple Grooves (3.4 cm x 3.4 cm): 34x34 mm
☐ Small Bare Surface (3.4 cm x 2.0 cm): _____ mm
☐ Large Bare Surface (3.4 cm x 4.0 cm): _____ mm
☐ Vessel Bare Surface (3.4 cm x 2 cm): _____ mm
☐ Porous Frits: _____ mm
Spacers: thinnest mm

Total Thickness-Bench: _____ mm

Material (Quartz, Granite, ...?): KH-6, D=665 μ m, Scraff
☒ Gouge - Particle Size, Size Distribution: 4125 μ m
☐ Bare Surface - Roughness, Grit: _____

Vert. Load Initial Value (kN): -0,06
Horz. Load Initial Value (kN): -0,15

Horizontal Load PID Setting:

gain: 0,007
Ti: 0,02
Td: 0,001

Normal Stress(es): _____ (kN) _____ (MPa)

Confining Pressure: _____ (MPa)
Pore Pressure SX: _____ (MPa)
Pore Pressure DX: _____ (MPa)

Sliding Velocity: 1 μ m/s

Acceleration: 10⁶ (5-10⁶) μ m/s²
Deceleration: "

Starting Position: _____
Shear Displ. Sample: _____

Velocity Steps: 10-3-10-70-100-300
Holds: 1-3-10-30-100-200-1000

Temperature (°C): RT
Relative Humidity (%): RH

Comments:

thinnest square B sample prep
position dropped during run - Par ~ 2s

604,85 614,79
605,47 615,20

HighSTEPS Experiment

Exp. Name: 127

Date: 17.5.22

Operator: Jonny Flann

Sample Holders Thickness w/o Gouge or Bare Surface:

☒ Simple Grooves (3.4 cm x 3.4 cm): 24 x 24 mm
☐ Small Bare Surface (3.4 cm x 2.0 cm): _____ mm
☐ Large Bare Surface (3.4 cm x 4.0 cm): _____ mm
☐ Vessel Bare Surface (3.4 cm x 2 cm): _____ mm
☐ Porous Frits: _____ mm
Spacers: _____ mm

Total Thickness-Bench: _____ mm

Material (Quartz, Granite, ...?): Sand, 10H-6 D=665 μ m

☒ Gouge - Particle Size, Size Distribution: <125 μ m

☐ Bare Surface - Roughness, Grit: _____

Vert. Load Initial Value (kN): -0.02

Horz. Load Initial Value (kN): -0.38

Horizontal Load PID Setting:

gain: 0.003

Ti: 0.02

Td: 0.001

Normal Stress(es): 11.56 (kN) 10 (MPa)

Confining Pressure: _____ (MPa)

Pore Pressure SX: _____ (MPa)

Pore Pressure DX: _____ (MPa)

Sliding Velocity: 10 μ m/s

Acceleration: 10⁰ (5-10⁰) μ m/s²

Deceleration: 4

Starting Position: _____

Shear Displ. Sample: _____

Velocity Steps: _____

Holds: _____

Temperature (°C): _____

Relative Humidity (%): _____

Comments:

Used small square for setup
Start filling 12:42 - 13:22
removed force / displac / shear displac drop at t=735s
(during walking)

$P_I = 604,90 \text{ g}$ $P_{I \text{ charge}} = 614,03 \text{ g}$

$P_{II} = 605,52 \text{ g}$ $P_{II \text{ charge}} = 614,65 \text{ g}$

HighSTEPS Experiment

Exp. Name: Barnaby Arthur K665 (h335)
Operator: Barnaby - Arthur

Date: 27.06.2022

Sample Holders Thickness w/o Gouge or Bare Surface:

___ Simple Grooves (3.4 cm x 3.4 cm): 34 x 34 mm
___ Small Bare Surface (3.4 cm x 2.0 cm): _____ mm
___ Large Bare Surface (3.4 cm x 4.0 cm): _____ mm
___ Vessel Bare Surface (3.4 cm x 2 cm): _____ mm
___ Porous Frits: _____ mm

Spacers: _____ mm

Total Thickness-Bench: _____ mm

Material (Quartz, Granite, ...?): altered basalt

___ Gouge - Particle Size, Size Distribution: < 125 μm

___ Bare Surface - Roughness, Grit: _____

Krafl KH-6 altered basalt $d = 665 \text{ nm}$

Vert. Load Initial Value (kN): 0.05 kN

Horz. Load Initial Value (kN): -0.13 kN

Horizontal Load PID Setting:

gain: 0.003

Ti: 0.02

Td: 0.001

Normal Stress(es): 34680 (kN) 30 (MPa)

Confining Pressure: _____ (MPa)

Pore Pressure SX: _____ (MPa)

Pore Pressure DX: _____ (MPa)

Sliding Velocity: 10 $\mu\text{m s}^{-1}$

Acceleration: _____

Deceleration: _____

Starting Position: _____

Shear Displ. Sample: _____

Velocity Steps: 3 - 10 - 30 - 100 - 300

Holds: 1 - 3 - 10 - 30 - 100 - 300 - 1000 - 3000

Temperature ($^{\circ}\text{C}$): RT

Relative Humidity (%): dry

Comments:

- thinness square
- Observation: ridges and color change (brown to gray)

$P_I = 604,87 \text{ g}$ $P_{I \text{ change}} = 613,92 \text{ g}$

$P_{II} = 605,47 \text{ g}$ $P_{II \text{ change}} = 614,35 \text{ g}$

HighSTEPS Experiment

Exp. Name: h336 (K665)
Operator: Barnaby - Arthur

Date: 28/06/2022

Sample Holders Thickness w/o Gouge or Bare Surface:

☐ Simple Grooves (3.4 cm x 3.4 cm): 34 x 34 mm
☐ Small Bare Surface (3.4 cm x 2.0 cm): _____ mm
☐ Large Bare Surface (3.4 cm x 4.0 cm): _____ mm
☐ Vessel Bare Surface (3.4 cm x 2 cm): _____ mm
☐ Porous Frits: _____ mm

Spacers: thinness _____ mm

Total Thickness-Bench: _____ mm

Material (Quartz, Granite, ...?): altered basalt

☐ Gouge - Particle Size, Size Distribution: < 125 μm

☐ Bare Surface - Roughness, Grit: _____

Krafl KH-6 altered basalt d = 665 m

Vert. Load Initial Value (kN): -0,03

Horz. Load Initial Value (kN): -0,15

Horizontal Load PID Setting:

gain: 0,003

Ti: 0,02

Td: 0,001

Normal Stress(es): 34680 (kN) 304 (MPa)

Confining Pressure: / (MPa)

Pore Pressure SX: / (MPa)

Pore Pressure DX: / (MPa)

Sliding Velocity: 10 $\mu\text{m s}^{-1}$

Acceleration: _____

Deceleration: _____

Starting Position: _____

Shear Displ. Sample: _____

Velocity Steps: 3-10-30-100-300

Holds: 1-3-10-30-100-300-1000-3000

Temperature ($^{\circ}\text{C}$): RT

Relative Humidity (%): saturated (time: 10h30 - 11h10)

Comments: - thinness square

606.13g 615.40g
606.58g 615.77g

HighSTEPS Experiment

Exp. Name: h433
Operator: Burnaby

Date: 3-11-23

Sample Holders Thickness w/o Gouge or Bare Surface:

☒ Simple Grooves (3.4 cm x 3.4 cm): 34 x 34 mm
☐ Small Bare Surface (3.4 cm x 2.0 cm): _____ mm
☐ Large Bare Surface (3.4 cm x 4.0 cm): _____ mm
☐ Vessel Bare Surface (3.4 cm x 2 cm): _____ mm
☐ Porous Frits: _____ mm

Spacers: thinnest mm

Total Thickness-Bench: _____ mm

Material (Quartz, Granite, ...?): k665

☒ Gouge - Particle Size, Size Distribution: <125 μ m
☐ Bare Surface - Roughness, Grit: _____

Vert. Load Initial Value (kN): -0.25

Horz. Load Initial Value (kN): -0.66

Horizontal Load PID Setting:

gain: 0.005

Ti: 0.02

Td: 0.001

Normal Stress(es): 57.8 (kN) 50 (MPa)

Confining Pressure: _____ (MPa)

Pore Pressure SX: _____ (MPa)

Pore Pressure DX: _____ (MPa)

Sliding Velocity: 10 μ m/s for 11mm

Acceleration: 566 μ m/s²

Deceleration: _____

Starting Position: _____

Shear Displ. Sample: _____

Velocity Steps: 10-3-10-30-100-300

Holds: 1-3-10-30-100-300-1000-2000

Temperature (°C): RT

Relative Humidity (%): dry

Comments:

thinnest square used for setup
magnets?

ops 614.62 g
604.87 g 614.00 g

HighSTEPS Experiment

Exp. Name: h434
Operator: Barnaby

Date: 4-11-22

Sample Holders Thickness w/o Gouge or Bare Surface:

☒ Simple Grooves (3.4 cm x 3.4 cm): 34 x 34 mm
☐ Small Bare Surface (3.4 cm x 2.0 cm): _____ mm
☐ Large Bare Surface (3.4 cm x 4.0 cm): _____ mm
☐ Vessel Bare Surface (3.4 cm x 2 cm): _____ mm
☐ Porous Frits: _____ mm
Spacers: thinnest mm

Total Thickness-Bench: _____ mm

Material (Quartz, Granite, ...?): h665
☒ Gouge - Particle Size, Size Distribution: < 125 μ m
☐ Bare Surface - Roughness, Grit: _____

Vert. Load Initial Value (kN): -0.2
Horz. Load Initial Value (kN): -0.5

Horizontal Load PID Setting:

gain: 0.003
Ti: 0.02
Td: 0.001

Normal Stress(es): 57.8 (kN) 50 (MPa)

Confining Pressure: _____ (MPa)
Pore Pressure SX: _____ (MPa)
Pore Pressure DX: _____ (MPa)

Sliding Velocity: 10 μ m/s for 11mm

Acceleration: 5e6 μ m/s²
Deceleration: _____

Starting Position: _____
Shear Displ. Sample: _____

Velocity Steps: 10-3-10-30-100-300
Holds: 1-3-10-30-100-300-1000-3000

Temperature (°C): RT
Relative Humidity (%): wet

Comments:
thinnest square used for setup
Saturated @ 1 MPa 8:43 - 9:24

sample holders 605, 49 614, 63
604, 88 613, 84

HighSTEPS Experiment

Exp. Name: h321
Operator: Barndt Florian

Date: 11.5.22

Sample Holders Thickness w/o Gouge or Bare Surface:

☒ Simple Grooves (3.4 cm x 3.4 cm): 34 x 34 mm
☐ Small Bare Surface (3.4 cm x 2.0 cm): _____ mm
☐ Large Bare Surface (3.4 cm x 4.0 cm): _____ mm
☐ Vessel Bare Surface (3.4 cm x 2 cm): _____ mm
☐ Porous Frits: _____ mm

Spacers: thin spacer _____ mm

Total Thickness-Bench: _____ mm

Material (Quartz, Granite, ...?): K708, KH-6, D=708um basalt

☒ Gouge - Particle Size, Size Distribution: < 125 μ m
☐ Bare Surface - Roughness, Grit: _____

Vert. Load Initial Value (kN): -0.15 / -0.175

Horz. Load Initial Value (kN): -0.4 / -0.44

Horizontal Load PID Setting:

gain: 0.003

Ti: 0.02

Td: 0.001

Normal Stress(es): 11.56 (kN) 10 (MPa)

Confining Pressure: _____ (MPa)

Pore Pressure SX: _____ (MPa)

Pore Pressure DX: _____ (MPa)

Sliding Velocity: 10 μ m/s

Acceleration: 10⁶ (5.10⁶ μ m/s²)

Deceleration: "

Starting Position: _____

Shear Displ. Sample: _____

Velocity Steps: 10 - 3 - 10 - 30 - 100 - 300 -

Holds: 1 - 3 - 10 - 30 - 100 - 300 - 1000 - 3000

Temperature (°C): RT

Relative Humidity (%): RH

Comments:

thinnest square used for sample prep
change of initial pore value: door closed

touch point: 7.5 mm

sample holders 604,87 613,80
605,46 614,37

HighSTEPS Experiment

Exp. Name: h 323
Operator: Barndy Florian

Date: 12.5

Sample Holders Thickness w/o Gouge or Bare Surface:

☒ Simple Grooves (3.4 cm x 3.4 cm): 34x34 mm
☐ Small Bare Surface (3.4 cm x 2.0 cm): _____ mm
☐ Large Bare Surface (3.4 cm x 4.0 cm): _____ mm
☐ Vessel Bare Surface (3.4 cm x 2 cm): _____ mm
☐ Porous Frits: _____ mm

Spacers: _____ mm

Total Thickness-Bench: _____ mm

Material (Quartz, Granite, ...?): KH-G, D = 708 m, basalt

☒ Gouge - Particle Size, Size Distribution: < 125 μ m
☐ Bare Surface - Roughness, Grit: _____

Vert. Load Initial Value (kN): -0.15
Horz. Load Initial Value (kN): -0.52

Horizontal Load PID Setting:

gain: 0.007
Ti: 0.02
Td: 0.001

Normal Stress(es): 11.50 (kN) 10 (MPa)

Confining Pressure: _____ (MPa)
Pore Pressure SX: _____ (MPa)
Pore Pressure DX: _____ (MPa)

Sliding Velocity: 10 μ m/s

Acceleration: 10⁶ (5.6⁶) μ m/s²
Deceleration: _____

Starting Position: _____
Shear Displ. Sample: _____

Velocity Steps: 10 - 3 - 10 - 30 - 100 - 300
Holds: 1 - 3 - 10 - 30 - 100 - 300 - 1000

Temperature (°C): RT
Relative Humidity (%): wet

Comments:

thinnest square
wetting 9424 -

$$P_I = 604,88g \quad P_{Ic} = 613,33g$$

$$P_{II} = 605,58g \quad P_{IIc} = 614,39g$$

HighSTEPS Experiment

Exp. Name: H339

Date: 26/07/2022

Operator: Florian - Arthur

Sample Holders Thickness w/o Gouge or Bare Surface:

___ Simple Grooves (3.4 cm x 3.4 cm): 34x34 mm

___ Small Bare Surface (3.4 cm x 2.0 cm): _____ mm

___ Large Bare Surface (3.4 cm x 4.0 cm): _____ mm

___ Vessel Bare Surface (3.4 cm x 2 cm): _____ mm

___ Porous Frits: _____ mm

Spacers: thinnest mm

Total Thickness-Bench: _____ mm

Material (Quartz, Granite, ...?): Krafla fresh basalt - KH-6 D = 708,5 m

☒ Gouge - Particle Size, Size Distribution: < 125 μ m

___ Bare Surface - Roughness, Grit: _____

Vert. Load Initial Value (kN): 0,018 N

Horz. Load Initial Value (kN): -0,35 kN

Horizontal Load PID Setting:

gain: 0,003

Ti: 0,02

Td: 0,001

Normal Stress(es): 34680 (kN) 30 (MPa)

Confining Pressure: _____ (MPa)

Pore Pressure SX: _____ (MPa)

Pore Pressure DX: _____ (MPa)

Sliding Velocity: 10 μ m.s⁻¹

Acceleration: _____

Deceleration: _____

Starting Position: _____

Shear Displ. Sample: _____

Velocity Steps: 3-10-30-100-300

Holds: 1-3-10-30-100-300-1000-3000

Temperature (°C): RT

Relative Humidity (%): dry

Comments:

thinnest square

$P_I = 605,54 \text{ g}$ $P_{Ic} = 614,36 \text{ g}$
 $P_{II} = 604,88 \text{ g}$ $P_{IIc} = 613,78 \text{ g}$

HighSTEPS Experiment

Exp. Name: H340
Operator: Florian - Arthuer

Date: 27/07/2022

Sample Holders Thickness w/o Gouge or Bare Surface:

☒ Simple Grooves (3.4 cm x 3.4 cm): 34x34 mm
___ Small Bare Surface (3.4 cm x 2.0 cm): _____ mm
___ Large Bare Surface (3.4 cm x 4.0 cm): _____ mm
___ Vessel Bare Surface (3.4 cm x 2 cm): _____ mm
___ Porous Frits: _____ mm

Spacers: thinest mm

Total Thickness-Bench: _____ mm

Material (Quartz, Granite, ...?): Krafla fresh basalt KH-6 D=708,5 mm

☒ Gouge - Particle Size, Size Distribution: <125 μm
___ Bare Surface - Roughness, Grit: _____

Vert. Load Initial Value (kN): 0,01 kN

Horz. Load Initial Value (kN): -0,23 kN

Horizontal Load PID Setting:

gain: 0,003

Ti: 0,02

Td: 0,001

Normal Stress(es): 34,680 (kN) 30 (MPa)

Confining Pressure: _____ (MPa)

Pore Pressure SX: _____ (MPa)

Pore Pressure DX: _____ (MPa)

Sliding Velocity: 10 $\mu\text{m.s}^{-1}$

Acceleration: _____

Deceleration: _____

Starting Position: _____

Shear Displ. Sample: _____

Velocity Steps: 1 - 3 - 10 - 30 - 100 - 300

Holds: 1 - 3 - 10 - 30 - 100 - 300 - 1000 - 3000

Temperature ($^{\circ}\text{C}$): RT

Relative Humidity (%): wet

(time: 10h55 - 11h36)

Comments:

606.15g 614.58g
606.56g 614.92g

HighSTEPS Experiment

Exp. Name: h435
Operator: Barnaby

Date: 4-11-22

Sample Holders Thickness w/o Gouge or Bare Surface:

☒ Simple Grooves (3.4 cm x 3.4 cm): 34 x 34 mm
☐ Small Bare Surface (3.4 cm x 2.0 cm): _____ mm
☐ Large Bare Surface (3.4 cm x 4.0 cm): _____ mm
☐ Vessel Bare Surface (3.4 cm x 2 cm): _____ mm
☐ Porous Frits: _____ mm

Spacers: thinnest mm

Total Thickness-Bench: _____ mm

Material (Quartz, Granite, ...?): h708

☒ Gouge - Particle Size, Size Distribution: <125 μ m
☐ Bare Surface - Roughness, Grit: _____

Vert. Load Initial Value (kN): -0.3

Horz. Load Initial Value (kN): -0.8

Horizontal Load PID Setting:

gain: 0.03

Ti: 0.07

Td: 0.001

Normal Stress(es): 57.8 (kN) 50 (MPa)

Confining Pressure: _____ (MPa)

Pore Pressure SX: _____ (MPa)

Pore Pressure DX: _____ (MPa)

Sliding Velocity: 10 μ m/s for 11mm

Acceleration: 5e6 μ m/s²

Deceleration: _____

Starting Position: _____

Shear Displ. Sample: _____

Velocity Steps: 10 - 30 - 10 - 30 - 100 - 200

Holds: 1 - 3 - 10 - 20 - 100 - 200 - 1000 - 300

Temperature (°C): RT

Relative Humidity (%): dry

Comments:

thinnest square used for setup

INITIAL WEIGHTS

WEIGHT w/ GOUGE

I = 605.53g

I = 614.12g

II = 604.88g

II = 613.19g

HighSTEPS Experiment

Exp. Name: h436
Operator: Barnaby

Date: 7-11-22

Sample Holders Thickness w/o Gouge or Bare Surface:

☒ Simple Grooves (3.4 cm x 3.4 cm): 34x34 mm
☐ Small Bare Surface (3.4 cm x 2.0 cm): _____ mm
☐ Large Bare Surface (3.4 cm x 4.0 cm): _____ mm
☐ Vessel Bare Surface (3.4 cm x 2 cm): _____ mm
☐ Porous Frits: _____ mm

Spacers: thinnest mm

Total Thickness-Bench: _____ mm

Material (Quartz, Granite, ...?): k708

☒ Gouge - Particle Size, Size Distribution: <125 μ m
☐ Bare Surface - Roughness, Grit: _____

Vert. Load Initial Value (kN): -0.2
Horz. Load Initial Value (kN): -0.24

Horizontal Load PID Setting:

gain: 0.003
Ti: 0.02
Td: 0.001

Normal Stress(es): 57.8 (kN) 50 (MPa)

Confining Pressure: _____ (MPa)
Pore Pressure SX: _____ (MPa)
Pore Pressure DX: _____ (MPa)

Sliding Velocity: 0 μ m/s for 11mm

Acceleration: 526 μ m/s²
Deceleration: _____

Starting Position: _____
Shear Displ. Sample: _____

Velocity Steps: 10-3-10-30-100-300
Holds: 1-3-10-30-100-300-1000-3000

Temperature (°C): RT
Relative Humidity (%): wet

Comments:

thinnest square used for ship
Saturated @ 1MPa 8:45- 9:25

sample holders: 604.85 612.12
605.47 614.00

HighSTEPS Experiment

Exp. Name: h328

Date: 14.6.22

Operator: Florian

Sample Holders Thickness w/o Gouge or Bare Surface:

☒ Simple Grooves (3.4 cm x 3.4 cm): 74 x 74 mm

☐ Small Bare Surface (3.4 cm x 2.0 cm): mm

☐ Large Bare Surface (3.4 cm x 4.0 cm): mm

☐ Vessel Bare Surface (3.4 cm x 2 cm): mm

☐ Porous Frits: mm

Spacers: thinnest mm

Total Thickness-Bench: mm

Material (Quartz, Granite, ...?): K771, Breccia, Kraft, ICH-6

☒ Gouge - Particle Size, Size Distribution: < 125 μ

☐ Bare Surface - Roughness, Grit:

Vert. Load Initial Value (kN): -0.075

Horz. Load Initial Value (kN): -0.27

Horizontal Load PID Setting:

gain: 0.007

Ti: 0.02

Td: 0.001

Normal Stress(es): 11.56 (kN) 10 (MPa)

Confining Pressure: (MPa)

Pore Pressure SX: (MPa)

Pore Pressure DX: (MPa)

Sliding Velocity: 10 μ /s

Acceleration:

Deceleration:

Starting Position:

Shear Displ. Sample:

Velocity Steps: 10 - 3 - 10 - 70 - 100 - 700

Holds: 1 - 3 - 10 - 70 - 100 - 700 - 1000 - 7000

Temperature (°C): RT

Relative Humidity (%): RH

Comments:

thinnest square sample prep 17.6.22
drops of some seconds in force & pos at t=7566s, 7618s,

sample holder 605,90 617,86 617,42
605,50 617,86

HighSTEPS Experiment

Exp. Name: h324
Operator: Brendy Florman

Date: 12.5

Sample Holders Thickness w/o Gouge or Bare Surface:

☒ Simple Grooves (3.4 cm x 3.4 cm): 34 x 34 mm
☐ Small Bare Surface (3.4 cm x 2.0 cm): _____ mm
☐ Large Bare Surface (3.4 cm x 4.0 cm): _____ mm
☐ Vessel Bare Surface (3.4 cm x 2 cm): _____ mm
☐ Porous Frits: _____ mm

Spacers: thinner _____ mm

Total Thickness-Bench: _____ mm

Material (Quartz, Granite, ...?): Breccia IC771 D=731 - FH-6

☒ Gouge - Particle Size, Size Distribution: <125 μ m
☐ Bare Surface - Roughness, Grit: _____

Vert. Load Initial Value (kN): -0.125

Horz. Load Initial Value (kN): -0.62964

Horizontal Load PID Setting:

gain: 0.003

Ti: 202

Td: 0.001

Normal Stress(es): 11.56 (kN) 10 (MPa)

Confining Pressure: _____ (MPa)

Pore Pressure SX: _____ (MPa)

Pore Pressure DX: _____ (MPa)

Sliding Velocity: 10 μ m/s

Acceleration: _____

Deceleration: _____

Starting Position: _____

Shear Displ. Sample: _____

Velocity Steps: 10-3-10-30-100-300

Holds: 1-3-10-30-100-300-1000-2000

Temperature (°C): RT

Relative Humidity (%): wet

Comments:

thinner square for sample prep
watting: 15403-15444

604,88 s 613,20 s
605,49 s 614,07 s

HighSTEPS Experiment

Exp. Name: h347
Operator: Florian

Date: 5.8.22

Sample Holders Thickness w/o Gouge or Bare Surface:

☒ Simple Grooves (3.4 cm x 3.4 cm): 34 x 34 mm
☐ Small Bare Surface (3.4 cm x 2.0 cm): _____ mm
☐ Large Bare Surface (3.4 cm x 4.0 cm): _____ mm
☐ Vessel Bare Surface (3.4 cm x 2 cm): _____ mm
☐ Porous Frits: _____ mm

Spacers: thinnest mm

Total Thickness-Bench: _____ mm

Material (Quartz, Granite, ...?): Kapla KH-6 D=731µ Breccia

☒ Gouge - Particle Size, Size Distribution: < 125 µm
☐ Bare Surface - Roughness, Grit: _____

Vert. Load Initial Value (kN): -0.04

Horz. Load Initial Value (kN): -0.12

Horizontal Load PID Setting:

gain: 0.003

Ti: 0.02

Td: 0.001

Normal Stress(es): 34.68 (kN) 30 (MPa)

Confining Pressure: _____ (MPa)

Pore Pressure SX: / (MPa)

Pore Pressure DX: _____ (MPa)

Sliding Velocity: 10 µm/s

Acceleration: 10⁶ (5.10⁶) µm/s²

Deceleration: 10⁶ (5.10⁶) µm/s²

Starting Position: _____

Shear Displ. Sample: _____

Velocity Steps: 10-3-10-30-100-300

Holds: 1-3-10-30-100-300-1000-3000 ~ 5000 s

Temperature (°C): RT

Relative Humidity (%): dry

Comments:

thinnest square for sample preparation
tape/scotch which may get between gouge & sample holder was cut out. on a test
last hold longer than 3000 s as software did not advance alone but restarted
counting after ~ 2900 s of hold



604,85 g 613,46 g
605,51 g 614,03 g

HighSTEPS Experiment

Exp. Name: 6346
Operator: Florian Bernaby / Alexander

Date: 4.8.22

Sample Holders Thickness w/o Gouge or Bare Surface:

☒ Simple Grooves (3.4 cm x 3.4 cm): 34 x 34 mm
☐ Small Bare Surface (3.4 cm x 2.0 cm): _____ mm
☐ Large Bare Surface (3.4 cm x 4.0 cm): _____ mm
☐ Vessel Bare Surface (3.4 cm x 2 cm): _____ mm
☐ Porous Frits: _____ mm

Spacers: thinnest mm

Total Thickness-Bench: _____ mm

Material (Quartz, Granite, ...?): breccia KH-6, D = 771 μ

☒ Gouge - Particle Size, Size Distribution: < 125 μ m
☐ Bare Surface - Roughness, Grit: _____

Vert. Load Initial Value (kN): + 0.02 kN

Horz. Load Initial Value (kN): - 0.32 kN

Horizontal Load PID Setting:

gain: 0.003

Ti: 0.02

Td: 0.001

Normal Stress(es): 34,68 (kN) 70 (MPa)

Confining Pressure: _____ (MPa)

Pore Pressure SX: _____ (MPa)

Pore Pressure DX: _____ (MPa)

Sliding Velocity: 10 μ m/s

Acceleration: 10⁶ (5.10⁶) μ m/s²

Deceleration: 10⁶ (5.10⁶) μ m/s²

Starting Position: _____

Shear Displ. Sample: _____

Velocity Steps: 10 - 3 - 10 - 70 - 100 - 300

Holds: 1 - 3 - 10 - 30 - 100 - 700 - 1000 - 7000

Temperature (°C): RT

Relative Humidity (%): saturated

Comments:

Sat 10:59 - 11:40

no suction when disassembling sample
large piece of scratch on one side of sample (~ 0.5 cm, in grooves)
→ clean data but different than 6324 (10 MPa = 0, saturated) consider to redo one of them

606.5g 614.15g
606.54g 614.57g

HighSTEPS Experiment

Exp. Name: K437
Operator: Burnaby

Date: 7-11-22

Sample Holders Thickness w/o Gouge or Bare Surface:

☒ Simple Grooves (3.4 cm x 3.4 cm): 34 x 34 mm
☐ Small Bare Surface (3.4 cm x 2.0 cm): _____ mm
☐ Large Bare Surface (3.4 cm x 4.0 cm): _____ mm
☐ Vessel Bare Surface (3.4 cm x 2 cm): _____ mm
☐ Porous Frits: _____ mm

Spacers: thinnest mm

Total Thickness-Bench: _____ mm

Material (Quartz, Granite, ...?): K731

☒ Gouge - Particle Size, Size Distribution: <125 μ m
☐ Bare Surface - Roughness, Grit: _____

Vert. Load Initial Value (kN): -0.27

Horz. Load Initial Value (kN): -0.47

Horizontal Load PID Setting:

gain: 0.003

Ti: 0.02

Td: 0.001

Normal Stress(es): 57.8 (kN) 50 (MPa)

Confining Pressure: / (MPa)

Pore Pressure SX: / (MPa)

Pore Pressure DX: / (MPa)

Sliding Velocity: 10 μ m/s for 11mm

Acceleration: 506 μ m/s²

Deceleration: _____

Starting Position: 3

Shear Displ. Sample: _____

Velocity Steps: 10-3-10-30-100-300

Holds: 1-3-10-30-100-300-1000-3000

Temperature (°C): RT

Relative Humidity (%): dry

Comments:

thinnest sphere used for setup

605.52g 613.42g
604.88g 612.68g

HighSTEPS Experiment

Exp. Name: n438
Operator: Bernady

Date: 8-11-22

Sample Holders Thickness w/o Gouge or Bare Surface:

☒ Simple Grooves (3.4 cm x 3.4 cm): 34 x 34 mm
☐ Small Bare Surface (3.4 cm x 2.0 cm): _____ mm
☐ Large Bare Surface (3.4 cm x 4.0 cm): _____ mm
☐ Vessel Bare Surface (3.4 cm x 2 cm): _____ mm
☐ Porous Frits: _____ mm
Spacers: thinnest mm

Total Thickness-Bench: _____ mm

Material (Quartz, Granite, ...?): k731
☒ Gouge - Particle Size, Size Distribution: <125µm
☐ Bare Surface - Roughness, Grit: _____

Vert. Load Initial Value (kN): -0.25
Horz. Load Initial Value (kN): -0.3

Horizontal Load PID Setting:

gain: 0.003
Ti: 0.02
Td: 0.001

Normal Stress(es): 57.8 (kN) 50 (MPa)

Confining Pressure: / (MPa)
Pore Pressure SX: / (MPa)
Pore Pressure DX: / (MPa)

Sliding Velocity: 10µm/s for 11mm

Acceleration: 5e6µm/s²
Deceleration: _____

Starting Position: _____
Shear Displ. Sample: _____

Velocity Steps: 10-3-10-20-100-300
Holds: 1-3-10-20-100-300-1000-2000

Temperature (°C): RT
Relative Humidity (%): wet

Comments:

thinnest square used for setup
Saturated @ 1MPa 9:12 - 9:43